

IRBEYSKOYE, Russian Federation

WMO: 295870

Lat: 55.6636N

Lon: 95.4975E

Elev: 252

StdP: 98.33

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.0	(c) -38.5	(d) -44.0	(e) 0.1	(f) -42.0	(g) -40.6	(h) 0.1	(i) -38.0	(j) 8.7	(k) -8.7	(l) 7.9	(m) -6.9	(n) 0.2	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.2	18.8	27.2	17.9	25.6	17.4	20.3	26.3	19.3	25.1	18.3	23.8	2.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.6	22.8	17.2	12.7	21.7	16.1	11.8	20.7	59.8	26.4	56.2	25.3	52.9	23.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.4	(c) 5.4	(e) -43.2	(f) 33.1	(g) 3.5	(h) 1.3	(i) -45.7	(j) 34.0	(k) -47.8	(l) 34.8	(m) -49.7	(n) 35.5	(o) -52.2	(p) 36.4		
(4) 7.4	(5) 6.4	(6) 5.4	(7) -42.9	(8) 22.1	(9) 2.9	(10) 1.4	(11) -45.1	(12) 23.1	(13) -46.8	(14) 23.9	(15) -48.4	(16) 24.7	(17) -50.5	(18) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-21.7	-16.8	-6.3	2.2	8.6	15.8	17.9	15.7	7.9	0.1	-11.4	-15.7		
	DBStd	14.83	10.19	8.60	6.95	4.07	4.69	3.99	2.56	3.24	4.08	5.03	9.30	9.50		
	HDD10.0	4401	982	752	505	237	85	5	0	1	87	308	641	798		
	HDD18.3	6828	1240	985	763	485	303	93	39	94	312	566	891	1056		
	CDD10.0	671	0	0	0	2	42	179	245	178	26	0	0	0		
	CDD18.3	58	0	0	0	0	1	18	26	13	0	0	0	0		
	CDH23.3	917	0	0	0	2	42	378	295	187	13	0	0	0		
	CDH26.7	214	0	0	0	0	6	104	61	43	1	0	0	0		
	WSAvg	1.9	1.6	1.8	2.2	2.5	2.5	1.8	1.6	1.5	1.8	2.1	2.1	2.0		
	PrecAvg	464	14	13	13	27	42	60	73	86	53	33	28	24		
	PrecMax	584	26	27	34	49	91	107	186	161	84	50	51	46		
(15) Precipitation	PrecMin	293	6	5	5	3	13	21	9	42	8	9	12	8		
	PrecStd	77	6	6	8	13	20	24	40	34	22	13	10	10		
	DB	1.6	3.2	10.9	20.9	27.6	31.7	31.0	30.4	25.5	15.7	8.7	2.7			
	MCWB	-1.1	0.0	5.0	10.2	14.8	19.6	20.0	20.1	16.1	8.4	4.6	-0.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-1.5	0.7	7.3	16.5	24.7	29.5	28.4	27.4	21.0	11.9	5.2	0.2			
	2% MCWB	-3.2	-1.8	2.9	8.4	13.6	18.3	18.9	19.2	14.2	7.2	2.4	-2.0			
	5% DB	-3.8	-1.0	5.2	13.5	21.6	27.4	26.6	25.3	18.0	9.2	3.0	-1.5			
	5% MCWB	-5.3	-3.0	1.6	6.6	12.1	17.7	18.6	17.8	12.7	5.6	0.8	-3.2			
	10% DB	-6.5	-3.2	3.3	10.8	19.0	25.5	24.9	23.2	15.2	7.1	0.5	-3.2			
	10% MCWB	-7.7	-4.7	0.4	5.4	11.1	16.9	17.9	16.8	11.2	4.3	-1.0	-4.5			
	0.4% WB	-0.9	0.1	5.5	10.9	16.4	21.2	21.8	21.6	16.3	9.7	5.1	0.2			
	0.4% MCDB	1.3	2.9	10.1	19.9	25.1	28.3	27.9	27.9	23.3	13.5	8.2	2.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-2.9	-1.6	3.3	9.0	14.7	19.7	20.4	19.9	14.8	7.7	2.6	-1.7			
	2% MCDB	-1.4	0.3	6.8	15.5	22.3	26.7	25.8	25.7	20.2	11.3	5.2	-0.1			
	5% WB	-5.2	-3.0	1.9	7.1	13.2	18.7	19.4	18.4	13.4	6.1	0.7	-3.1			
	5% MCDB	-3.9	-1.2	4.8	12.5	20.0	25.2	24.6	24.0	17.4	8.8	2.5	-1.6			
	10% WB	-7.7	-4.9	0.5	5.5	11.7	17.6	18.5	17.4	11.5	4.6	-0.8	-4.4			
	10% MCDB	-6.4	-3.1	2.9	10.5	17.9	23.9	23.6	22.3	14.7	6.9	0.4	-3.1			
	MDBR	11.5	13.6	13.1	11.3	12.6	13.8	11.3	11.5	11.1	9.5	9.7	10.0			
	5% DB	11.8	13.6	14.6	17.9	18.8	18.9	15.6	16.6	17.7	13.9	9.7	9.0			
(36) Mean Daily Temperature Range	5% MCWBR	10.3	11.3	10.3	10.4	9.3	8.5	7.1	8.3	10.3	9.1	6.7	7.7			
	5% WB	11.1	12.7	13.2	16.1	15.9	16.0	13.2	15.1	16.1	13.2	9.7	8.6			
	5% MCWBR	9.9	10.7	9.5	10.0	8.3	7.8	6.7	8.0	9.7	9.0	7.5	7.4			
	Clear-Sky Solar Irradiance	taub	0.279	0.314	0.356	0.394	0.414	0.409	0.428	0.407	0.365	0.346	0.312	0.281		
(40) Clear-Sky Solar Irradiance	taud	2.028	2.004	1.981	2.134	2.183	2.296	2.262	2.288	2.356	2.257	2.090	1.973			
	Ebn at Noon	596	720	784	819	826	837	809	801	788	693	547	476			
	Edh at Noon	66	101	133	133	134	122	124	113	92	79	63	56			
(44) All-Sky Solar Radiation	RadAvg	0.82	1.83	3.27	4.35	4.95	5.93	5.35	4.33	2.81	1.55	0.79	0.51			
	RadStd	0.08	0.11	0.23	0.24	0.49	0.51	0.41	0.37	0.32	0.12	0.09	0.05			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page